

Wannes Fraj

Edge IA Engineer

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SUMMARY

Industrial Computer Engineering student at ENET'Com, focused on robotics, Edge AI, and embedded systems. Experienced with **ROS 2**, autonomous navigation, computer vision, deep learning, and IoT, with hands-on work on real-time perception and intelligent robotic systems. Familiar with embedded development and AI deployment for connected and autonomous applications.

EDUCATION

- **National School of Electronics and Telecommunications of Sfax(ENET'Com)** 2025 – Present
Industrial Computer Engineering Sfax, Tunisia
- **Faculty of Sciences of Monastir** 2023 – 2025
Preparatory Cycle in Physics and Chemistry Monastir, Tunisia

EXPERIENCE

- **Enova Robotics** June – July 2026
Deep Learning & Computer Vision Intern Sousse, Tunisia
 - Designed and implemented a full **ROS2**-based intelligent person-following system, structured into four communicating nodes: camera stream, person detector, distance estimator, and follow controller.
 - Built a custom **deep learning** dataset from self-captured photos to train a **YOLOv8** model for real-time person detection with **ByteTrack**, feeding detections into **Depth Anything V2** for distance estimation.
 - Identified the target person through facial recognition with **DeepFace** (FaceNet), fusing computer vision and deep learning into one real-time **ROS2** perception pipeline.
- **Tunipage — NA Consulting** July 2026 – Present
AI & DevOps Intern Monastir, Tunisia
 - Redesigned the **backend architecture**, API layer and data flow to fix a broken **OpenAI API** integration and make the **Django** AI feature **scalable** and **reliable** in production, eliminating server crashes under load.
 - Owned deployment end to end: **Docker** images, **Kubernetes** orchestration, and **GitLab CI/CD** pipelines.

TECHNICAL STACK

- **Languages:** Python, C, C++, Java, JavaScript, MATLAB
- **Edge AI & Computer Vision:** PyTorch, TensorFlow, Scikit-learn, YOLOv8
- **Robotics & Navigation:** ROS 2, Nav2, SLAM, LiDAR, RViz, Gazebo
- **Embedded Systems & IoT:** ESP32, MQTT, Mosquitto, Motor Control, Kafka
- **Deployment & Development:** Docker, Kubernetes, FastAPI, MCP, GitLab CI/CD, Git, Linux
- **Data & Simulation:** NumPy, Pandas, Matplotlib, Jupyter, Proteus 8, Feature Engineering

PROJECTS

- **Smart Home — Wireless Roller Shutter System** 2026
Tools: ESP32, C++, MQTT, BLE, Flutter, PlatformIO [\[🔗\]](#)
 - Built ESP32 firmware and a Flutter application for real-time shutter control using **MQTT**, **BLE**, and manual override.
- **CubeSat Thermal Prediction — Embedded AI** 2025
Tools: Python, LSTM, Operational Research [\[🔗\]](#)
 - Trained an **LSTM** to predict panel orientations for thermal risk management and autonomous CubeSat decision-making.
- **EventPlanner — Agentic AI Chatbot (2nd Place — N8N Hackathon)** 2026
Tools: N8N, LLMs, Agentic Workflows, REST APIs [\[🔗\]](#)
 - Built an agentic chatbot automating event planning through multi-step LLM orchestration, intent recognition, and real-time API retrieval.

CERTIFICATIONS

- **Deep Learning – NVIDIA DLI** 2026

VOLUNTEERING

- **Webmaster** Jan 2026 – Present
IEEE ENET'Com WIE Student Affinity Group
 - Managed the group's online presence, publication schedules, and digital communication for workshops and events.